

Larsson-Fibonacci Trading System

Complete Trading Rules & Logic Documentation

System Overview

The Larsson-Fibonacci Trading System combines two powerful technical analysis methods:

1. **Larsson Line** - A trend-following system using 4 RMA (Running Moving Average) periods to identify trend strength and direction
2. **Fibonacci Retracements** - Entry zones calculated from swing high/low over 100-day lookback period

This system is designed for **trend-following entries with precise risk management**, focusing on Weekly timeframe trend confirmation with Daily timeframe execution.

Larsson Line Logic

Four RMA Periods

- **V1 (Fast)**: 15-period RMA
- **M1**: 19-period RMA
- **M2**: 25-period RMA
- **V2 (Slow)**: 29-period RMA

Trend Status (Weekly & Daily)

GOLD (Bullish - Perfect Alignment)

- All 4 lines aligned in ascending order
- Condition: $V1 > V2$ AND $V1 > M1$ AND $M1 > M2$ AND $M2 > V2$
- Visual: Yellow/Gold color
- Interpretation: Strong bullish trend, all moving averages aligned

BLUE (Bearish - Perfect Reversal)

- All 4 lines aligned in descending order
- Condition: $V1 < V2$ AND $V1 < M1$ AND $M1 < M2$ AND $M2 < V2$
- Visual: Blue color
- Interpretation: Strong bearish trend, complete reversal

GRAY (Neutral - Mixed Alignment)

- Lines are mixed (neither all ascending nor descending)
- Condition: Does not meet GOLD or BLUE criteria

- Visual: Gray color
 - Interpretation: Weakening trend, consolidation, or transition phase
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Fibonacci Retracement Levels

Calculation Method

- **Lookback Period:** 100 days (on Daily timeframe)
- **Swing High:** Highest high over lookback period
- **Swing Low:** Lowest low over lookback period
- **Formula:** $\text{Level} = \text{Swing High} - (\text{Swing High} - \text{Swing Low}) \times \text{Percentage}$

Three Key Levels

1. **38.2%** - Shallowest retracement (strong trend continuation)
 2. **50.0%** - Mid-level retracement (balanced support)
 3. **61.8%** - Golden ratio (deepest acceptable pullback)
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Entry Rules

Two Entry Types

1. *PULLBACK Entry (Strong Buy Signal)*

Direction: Price crossing DOWN from above Fibonacci level

Detection Logic:

- Previous bar was ABOVE the Fib level
- Current price is within 2% tolerance of the Fib level
- Example: Stock pulls back from \$150 to \$145 (50% Fib level)

Position Sizing:

- 38.2% Level: **20% of capital**
- 50.0% Level: **30% of capital**
- 61.8% Level: **50% of capital**

Why Strong: Price respected the level from above, showing support

2. RECOVERY Entry (Confirmation Required)

Direction: Price crossing UP from below Fibonacci level

Detection Logic:

- Previous bar was BELOW the Fib level
- Price must cross ABOVE a **confirmation buffer** (+1.5% above the Fib level)
- Example: 50% Fib at \$145 requires price to reach \$147.18 for confirmation

Position Sizing:

- 38.2% Level: **20% of capital**
- 50.0% Level: **30% of capital**
- 61.8% Level: **50% of capital**

Why Buffer Required: Prevents false signals when price just touches the level from below without real strength

Complete BUY Conditions (All Required)

1. **Weekly Larsson = GOLD** (Strong bullish trend on Weekly timeframe)
2. **Price Above Weekly 200 SMA** (Long-term weekly support confirmed)
 - *Note: For stocks without 200 weeks of history (NaN), this requirement is waived*
3. **Price Above Daily 200 SMA** (Daily support confirmed)
4. **Price At Fibonacci Level** (Either pullback OR recovery with confirmation)
5. **Volume Declining** (3-period volume SMA < 20-period volume SMA)

When ALL conditions met: Dashboard shows "**BUY**" in bright green with entry details

Exit Rules

EXIT 50% (Partial Exit)

Trigger: Weekly Larsson turns **GRAY**

Logic:

- Trend is weakening but not yet reversed
- Take profits on 50% of position
- Keep 50% in case trend resumes

Dashboard Display: Orange background, "EXIT 50%"

EXIT 100% (Full Exit)

Triggers (Either condition):

1. Weekly Larsson turns **BLUE** (bearish reversal)
2. Price breaks below **Weekly 200 SMA**

Logic:

- Trend has reversed OR long-term support broken
- Exit entire position to preserve capital

Dashboard Display: Red background, "EXIT 100%"

Signal Types

1. BUY

- **Color:** Bright Green (Lime)
 - **Meaning:** All 5 conditions met, enter position
 - **Action:** Buy at current Fibonacci level with specified allocation
 - **Details Shown:**
 - Which Fibonacci level triggered (38.2%, 50%, 61.8%)
 - Entry type (Pullback or Recovery)
 - Position size percentage
 - Stop loss price and risk %
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2. WAIT

- **Color:** Yellow (Semi-transparent)
 - **Meaning:** Weekly Larsson is GOLD but other conditions not met
 - **Common Reasons:**
 - Price not at Fibonacci level yet
 - Volume not declining (still rising)
 - Price below Daily 200 SMA
 - **Action:** Monitor, do not enter yet
 - **Note:** Green arrow () may appear next to Fibonacci level showing proximity
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3. EXIT 50%

- **Color:** Orange
- **Trigger:** Weekly Larsson = GRAY
- **Action:** Sell 50% of position, hold 50%

4. EXIT 100%

- **Color:** Red
 - **Trigger:** Weekly Larsson = BLUE OR price below Weekly 200 SMA
 - **Action:** Sell entire position immediately
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5. NO SETUP

- **Color:** Yellow (Semi-transparent)
 - **Meaning:** Weekly Larsson is not GOLD (either GRAY or BLUE)
 - **Action:** No trade opportunity, stay in cash
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Volume Analysis

Purpose

Confirms that selling pressure is exhausting before entry

Calculation

- **Short-term Average:** 3-period SMA of volume
- **Long-term Average:** 20-period SMA of volume

Declining Volume (Required for BUY)

- **Condition:** 3-period SMA < 20-period SMA
- **Dashboard:** "Declining " in green
- **Interpretation:** Recent volume lower than average, selling exhaustion

Rising Volume (Blocks BUY)

- **Condition:** 3-period SMA >= 20-period SMA
 - **Dashboard:** "Rising " in red
 - **Interpretation:** Active selling still occurring, wait for exhaustion
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200 SMA Requirements

Daily 200 SMA

- **Calculation:** 200-period Simple Moving Average on **Daily** timeframe
- **Required:** Price must be ABOVE this level
- **Purpose:** Confirms intermediate-term uptrend
- **Dashboard Display:** "D:+10.3%" (example: 10.3% above)

Weekly 200 SMA

- **Calculation:** 200-period Simple Moving Average on **Weekly** timeframe
 - **Required:** Price must be ABOVE this level (or NaN for newer stocks)
 - **Purpose:** Confirms long-term uptrend
 - **Special Rule:** If NaN (stock doesn't have 200 weeks of history), requirement is automatically passed
 - **Dashboard Display:** "W:+227.5%" (example: 227.5% above)
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Stop Loss Calculation

Formula

$\text{Stop Loss} = \text{MAX}(\text{Daily 200 SMA} \times 0.90, 61.8\% \text{ Fib Level} \times 0.93)$

Components

1. **Daily 200 SMA × 0.90:** 10% below Daily 200 SMA
2. **61.8% Fib × 0.93:** 7% below the deepest Fibonacci level

Selection

Use whichever is HIGHER (less aggressive stop)

Risk Calculation

$\text{Risk \%} = ((\text{Current Price} - \text{Stop Loss}) / \text{Current Price}) \times 100$

Displayed in dashboard when BUY signal appears

Dashboard Indicators

Larsson Status Row

Format: W: **GOLD** | D: **GRAY**

- **W:** Weekly Larsson status (primary trend)
 - **D:** Daily Larsson status (entry timing)
 - **:** Appears only when Weekly is GOLD
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200 SMA Row

Format: W: **+227.5%** | D: **+10.3%**

- **W:** Distance from Weekly 200 SMA
 - **D:** Distance from Daily 200 SMA
 - **+**: Above SMA (bullish)
 - **-**: Below SMA (bearish)
 - **:** Price is above SMA
 - **:** Price is below SMA
 - **NaN:** Insufficient data (newer stocks)
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Volume Row

Format: **Declining** or **Rising**

- **Declining:** 3-period < 20-period (ready for entry)
 - **Rising:** 3-period >= 20-period (wait)
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Fibonacci Levels

Format: **38.2%** (**20%**): **\$152.16** (**+43.8%**)

- **38.2%:** Fibonacci level percentage
 - **(20%):** Position allocation if triggered
 - **\$152.16:** Actual price level
 - **(+43.8%):** Distance from current price (+ = above, - = below)
 - **:** Green arrow indicates current price is AT this level
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Trading Workflow

1. Check Weekly Larsson

- If not GOLD → **NO SETUP** → Stay in cash
- If GOLD → Continue evaluation

2. Check 200 SMAs

- Weekly: Must be above (or NaN for new stocks)
- Daily: Must be above
- If either fails → **WAIT**

3. Check Fibonacci Levels

- Is price at any of the three levels?
- Green arrow () indicates proximity
- If not at level → **WAIT**

4. Check Volume

- Is volume declining (3-period < 20-period)?
- If rising → **WAIT**
- If declining → **BUY** signal appears

5. Execute Entry

- Note the Fibonacci level hit (38.2%, 50%, 61.8%)
- Check entry type (Pullback or Recovery)
- Use specified position size (20%, 30%, or 50%)
- Set stop loss at calculated price

6. Monitor Position

- **Weekly Larsson turns GRAY** → Exit 50%
 - **Weekly Larsson turns BLUE** → Exit 100%
 - **Price breaks Weekly 200 SMA** → Exit 100%
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Key Strategy Principles

1. Weekly Trend is King

- Only take trades when Weekly Larsson = GOLD
- Weekly trend determines position direction
- Daily timeframe only for entry timing

2. Fibonacci Levels are Entry Zones

- Not exact prices, but zones with tolerance
- Deeper pullbacks = larger position sizes
- Recovery entries need confirmation buffer

3. Volume Confirms Entry

- Wait for volume to decline before entry
- Rising volume = continued selling pressure
- Declining volume = exhaustion, ready to bounce

4. Risk Management is Mandatory

- Stop loss calculated automatically
- Based on Daily 200 SMA and 61.8% Fib
- Risk % displayed before entry

5. Partial Exits Preserve Capital

- GRAY = weakening trend, take 50% profit
 - BLUE = reversal, exit completely
 - Never hold through Weekly BLUE
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Common Scenarios Explained

Scenario 1: "Why WAIT when at Fibonacci level?"

Answer: Check other conditions:

- Volume may be rising (not declining yet)
- Price may be below Daily 200 SMA
- Weekly 200 SMA may be broken

All 5 conditions must be met for BUY signal.

Scenario 2: "Green arrow but no BUY signal?"

Answer: Green arrow () only indicates price is AT a Fibonacci level. Other conditions (Weekly GOLD, volume declining, above 200 SMAs) must also be met.

Scenario 3: "EXIT 100% but Weekly is GOLD?"

Answer: Check Weekly 200 SMA. If price broke below it, EXIT 100% triggers regardless of Larsson status. Long-term support is broken.

Scenario 4: "Should I add to position at deeper Fib level?"

Answer: System allocates progressively:

- First entry: 20% at 38.2%
 - If continues down: 30% more at 50%
 - If continues down: 50% more at 61.8%
 - Total: 100% allocated across three levels
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Scenario 5: "Stock is new, showing NaN for Weekly SMA?"

Answer: System automatically passes Weekly 200 SMA requirement for newer stocks (like ALAB). Can still generate BUY signals based on Weekly Larsson GOLD and other conditions.

Visual Indicators on Chart

Fibonacci Lines

- **Green Lines:** Price is ABOVE the level (support held)
- **Red Lines:** Price is BELOW the level (support broken)
- **Labels:** Show percentage and exact price
- **Line Width:** Adjustable in settings

200 SMA Lines (Optional Display)

- **Red Line:** Daily 200 SMA
- **Black Line:** Weekly 200 SMA
- Toggle on/off in Display settings

Larsson Bands (Cloud)

- **Yellow Cloud:** Larsson lines on chart (optional)
- **Blue Cloud:** Bearish Larsson alignment
- Shows the RMA band structure visually

Settings & Customization

Larsson Line Settings

- V1 (Fast): Default 15
- M1: Default 19
- M2: Default 25
- V2 (Slow): Default 29

Fibonacci Settings

- Lookback Period: Default 100 days
- Fib Tolerance: Default 2% (for pullback detection)
- Recovery Buffer: Default 1.5% (above level for confirmation)
- Manual Override: Can set manual swing high/low

Position Sizing

- **Pullback Allocations:** 20% / 30% / 50%
- **Recovery Allocations:** 20% / 30% / 50%
- Customizable per level

Volume Settings

- Short Period: Default 3
- Long Period: Default 20

Display Settings

- Dashboard Position: Top Right / Top Left / Bottom Right / Bottom Left
 - Dashboard Size: Small / Normal / Large
 - Show Fibonacci Lines: On/Off
 - Show 200 SMA Lines: On/Off
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Alert Configuration

Available Alerts

1. **BUY Signal** - All 5 conditions met
2. **EXIT 50%** - Weekly Larsson turned GRAY
3. **EXIT 100%** - Weekly BLUE or Weekly 200 SMA broken

4. **Approaching Fib 38.2%** - Within 3% of level
5. **Approaching Fib 50.0%** - Within 3% of level
6. **Approaching Fib 61.8%** - Within 3% of level

Setup in TradingView

1. Click "Alert" button
 2. Select "Larsson-Fibonacci Trading Dashboard v2"
 3. Choose condition from dropdown
 4. Set notification preferences
 5. Name the alert
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Quick Reference Card

BUY Checklist (All Required)

Weekly Larsson = GOLD
Above Weekly 200 SMA (or NaN)
Above Daily 200 SMA
At Fibonacci Level (green)
Volume Declining ()

Exit Triggers

- **50%:** Weekly → GRAY
- **100%:** Weekly → BLUE OR break Weekly 200 SMA

Position Sizing

- **38.2%:** 20%
- **50.0%:** 30%
- **61.8%:** 50%

Entry Types

- **Pullback:** From above (strong)
 - **Recovery:** From below (needs +1.5% buffer)
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Risk Disclaimer

This trading system is for educational purposes. Past performance does not guarantee future results. Always:

- Use proper position sizing

- Set stop losses
 - Never risk more than you can afford to lose
 - Consider your risk tolerance
 - Consult a financial advisor
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System Name: Larsson-Fibonacci Trading Dashboard v2

Platform: TradingView (PineScript v5)